

# **Successful and inexpensive DNA extraction protocol from insects with low amount of tissue**

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## **Protocol:**

Considering that the insect to be used for DNA extraction was preserved/collected in Absolute Alcohol (Molecular grade). Dry the sample and transfer into the new microcentrifuge tube.

1. Add pre-warmed CTAB (600ul) (65°C) +3ul of Beta-Mercaptoethanol+ 10ul of 20% SDS
2. Crush the sample and then add 3ul of Proteinase-K.
3. Vortex each tube for 3-5 minutes vigorously.
4. Incubate at 65 °C for 4-5 hours (prefer shaking heating block), or vortex it at the interval of 30 minutes if possible.
5. After incubation, cool the samples to room temperature, and centrifuge at 14,000 rpm for 10 min
6. Add equal volume of Phenol: chloroform: Isoamyl alcohol (25:24:1).
7. Vortex for 3-5 minutes and then centrifuged at 14,000 rpm for 10 minutes.
8. Take supernatant in fresh microfuge tube and discard the pellet.
9. Add Chloroform: Isoamyl alcohol (24:1), 600ul in each tube and mix it (vortex).
10. Centrifuge at 12,000 rpm for 10 minutes.
11. Take the supernatant in fresh microcentrifuge tube.
12. Add chilled isopropanol (400 ul) and mixed slowly until white flakes appear.
13. Then keep it at -20°C in deep freeze for 1hour.
14. Bring the sample at room temperature and then centrifuge at 10,000 rpm for 10 minutes.

15. Decant the supernatant and add 70% chilled ethanol (400ul) + ammonium acetate (100 ul) to the pellet for washing.
16. Centrifuge at 10,000 rpm for 10 minutes.
17. Decant the supernatant carefully; add 400ul of absolute alcohol and Centrifuge at 10,000 rpm for 10 minutes.
18. Decant the supernatant and dry the pellet at room temperature or in speed vac.  
*Speed vac program: Program name '-OH', for 15 minutes at 30°C.*
19. Add nuclease free water (10-30ul) and incubate samples in a dry bath at 55°C for 30 minutes.
20. Keep/store at -20°C or -80°C (long term storage)

#### CTAB EXTRACTION BUFFER – (For 50ml)

1M Tris-HCl 5ml, 0.5 M EDTA 2ml, 5M NaCl 17.5ml, 10% CTAB 10ml (5 gm in 50 ml), Distilled Water 15.5 ml to make up the volume. Set the pH to 7.5-8.0. (Autoclave the whole content before use.)